Mathematical Modeling of Cigarette and E-Cigarette Dynamics: A Delay Differential Equations Approach

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Code: github.com/revak18/nicotine_use_mathmodel (NicotineUse_SIR_DDE_Model_v2.ipynb)

Revised version. An earlier version reported a close match between the model and 2023 CDC data. That match came from an error in the comparison code; this version corrects it and reports what the corrected comparison shows.

Abstract

This study uses an age-structured system of nine delay differential equations (DDEs) to model cigarette and e-cigarette use among youth (ages 11–17), young adults (18–23), and adults (24+). Each age group is divided into susceptible, current-user, and former-user states, with delays of 12 months for initiation, 6 months for quitting, and 12 months for relapse. The model is initialized with 2022 CDC data and simulated for 120 months.

Comparing the model's predictions with 2023 and 2024 National Youth Tobacco Survey data showed that the original setup greatly overpredicts youth e-cigarette use: 53.5% predicted versus 7.7% observed in 2023. Three assumptions explain most of the gap: starting from the share of youth who had ever used e-cigarettes instead of current users, a birth rate that does not balance aging, and treating an 18% offer-acceptance figure as a monthly transmission rate. With all three corrected, the model comes close to the CDC values in both years (7.8% vs. 7.7% in 2023, 6.8% vs. 5.9% in 2024), though no single transmission rate matches both years exactly.

Because transmission enters the equations only as the product of contact frequency and acceptance probability, prevalence data cannot identify these two factors separately. For cigarettes, predictions depend mainly on the starting prevalence and the quit rate and

barely respond to the transmission rate. The results show what an age-structured delay model can capture, and what population-level prevalence data can and cannot determine about the mechanisms behind nicotine use.

1. Introduction

Nicotine use remains one of the leading public health challenges worldwide. Tobacco use accounts for more than 8 million deaths each year and more than $1.4 trillion in healthcare costs and lost productivity (World Health Organization, 2023).

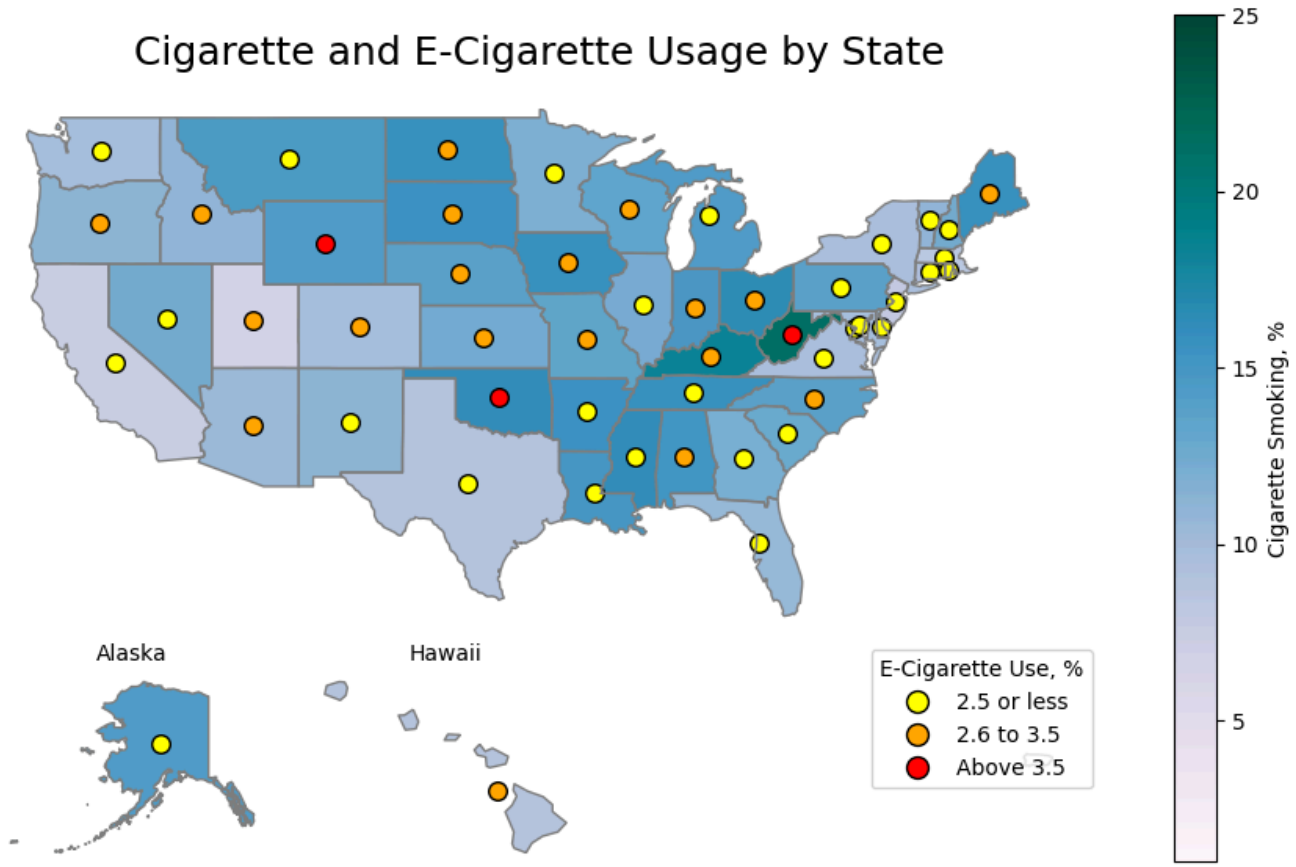


Figure 1. Cigarette and e-cigarette use by state (data: CDC).

Traditional cigarette use has declined in recent decades, but e-cigarettes have changed the picture. They are marketed as less harmful alternatives and have spread quickly among adolescents and young adults, who are strongly influenced by marketing, social media, and peers. E-cigarette aerosols contain toxic substances, and their nicotine delivery produces dependence comparable to smoking (National Institute on Drug Abuse, 2025). In 2022, 14.1% of U.S. high school students and 3.3% of middle school students reported current e-cigarette use, far more than reported cigarette use (Cooper et al., 2022), and 6.0% of adults used e-cigarettes (CDC, 2022).

Adolescents are especially vulnerable, because nicotine exposure during development can cause long-term cognitive and behavioral effects (Yuan et al., 2015). Relapse is also common: nearly 43% of people resume nicotine use within a year of quitting (U.S. Surgeon General, 2023).

Susceptible–Infected–Recovered (SIR) models from epidemiology have been adapted to study behaviors such as substance use. Chao et al. (2019) used agent-based modeling to study social influence on smoking and vaping among high school students, and Wagner and Clifton (2021) modeled competition between cigarettes and e-cigarettes. Most such models use ordinary differential equations, in which a change in one group affects others instantly. Starting, quitting, and relapsing, however, unfold over months. Delay differential equations (DDEs) represent these lags directly. Existing models also tend to study youth or adults separately, although people move from one age group to the next.

This study:

- builds a nine-equation DDE model of nicotine use across youth, young adults, and adults, with delays for initiation (τ_1), quitting (τ_2), and relapse (τ_3);
- simulates cigarettes and e-cigarettes separately, with product-specific transmission (β), quitting (γ), and relapse (ρ) rates;
- compares the model's predictions with 2023 and 2024 CDC survey data, identifies which assumptions drive the differences, and asks which parameters the data can actually determine.

2. Methodology

2.1 Assumptions

1. **Nicotine use spreads like an infection.** Susceptible people start using only through contact with current users (peer influence), consistent with the description of youth vaping as an epidemic since 2018.
2. **Age range.** Use can begin at age 11 and continue for life.
3. **Births and deaths.** New 11-year-olds enter the youth group at a constant rate $B = 0.00103$ per month. Everyone dies at the same rate $\mu = 0.0006657$ per month.
4. **Three states per age group.** Susceptible (S), current user (I), and former user (R).
5. **Time-varying rates.** Transmission (β), quitting (γ), and relapse (ρ) vary slightly over time (Section 2.3).
6. **No former users at the start.** $R = 0$ in every group in 2022.

7. **One-way age flow.** Youth age into young adults, and young adults into adults. Young adults start using through contact with youth users (Section 2.5).

2.2 Data and initial conditions (2022)

Youth (ages 11–17). From the 2022 National Youth Tobacco Survey (Figure 2), 20.0% of middle and high school students reported *ever* using e-cigarettes and 8.3% reported ever using cigarettes. The model's youth starting values are therefore $I_0 = 0.20$ for e-cigarettes and $I_0 = 0.083$ for cigarettes, with $S_0 = 1 - I_0$ and $R_0 = 0$. Section 4 shows that using ever-use rather than current use is one of the main sources of error.

LEMENTARY TABLE. Percentage of middle and high school students who reported ever using tobacco product, by product,* overall and by school level, sex, and race and city — National Youth Tobacco Survey, United States, 2022

Tobacco Product	Sex		Race and ethnicity†							Total	
	Female, % (95% CI)	Male, % (95% CI)	White, NH, % (95% CI)	Black or African American, NH, % (95% CI)	Hispanic, % (95% CI)	Asian, NH, % (95% CI)	AI/AN, NH, % (95% CI)	NH/OPI, NH, % (95% CI)	Multiple, NH, % (95% CI)	% (95% CI)	Es e u
Overall											
Any tobacco product¶	25.7 (23.4–28.2)	24.0 (21.5–26.7)	26.4 (23.6–29.5)	24.6 (21.4–28.2)	24.9 (22.9–27.0)	11.4 (7.8–16.5)	30.2 (23.9–37.3)	21.6 (12.3–35.2)	28.7 (23.9–34.1)	**24.8 (22.5–27.3)**	6,
E-cigarettes	21.6 (19.3–24.1)	18.6 (16.2–21.2)	22.2 (19.4–25.3)	16.7 (13.9–19.8)	20.2 (18.4–22.3)	7.5 (4.4–12.7)	19.6 (14.8–25.5)	—**	23.8 (19.2–29.1)	**20.0 (17.8–22.4)**	5,
Cigarettes	8.2 (7.0–9.5)	8.5 (7.1–10.1)	9.8 (8.2–11.6)	5.5 (4.3–6.9)	8.1 (6.9–9.3)	3.1 (2.3–4.1)	15.5 (11.0–21.5)	—	8.4 (6.3–11.0)	**8.3 (7.2–9.6)**	2,
Cigars	4.3 (3.6–5.2)	7.1 (5.8–8.7)	6.3 (5.1–7.8)	7.3 (5.7–9.1)	4.7 (3.8–5.8)	—	7.6 (4.4–12.9)	—	7.1 (5.2–9.7)	**5.7 (4.8–6.9)**	1,

Figure 2. Percentage of middle and high school students who reported ever using tobacco products, National Youth Tobacco Survey, 2022 (Credit: CDC).

Young adults and adults (18+). From the 2022 National Health Interview Survey (Figure 3), 6.0% of adults used e-cigarettes and 11.6% smoked cigarettes. Both adult groups start at $I_0 = 0.06$ (e-cigarettes) or $I_0 = 0.116$ (cigarettes), with $S_0 = 1 - I_0$ and $R_0 = 0$.

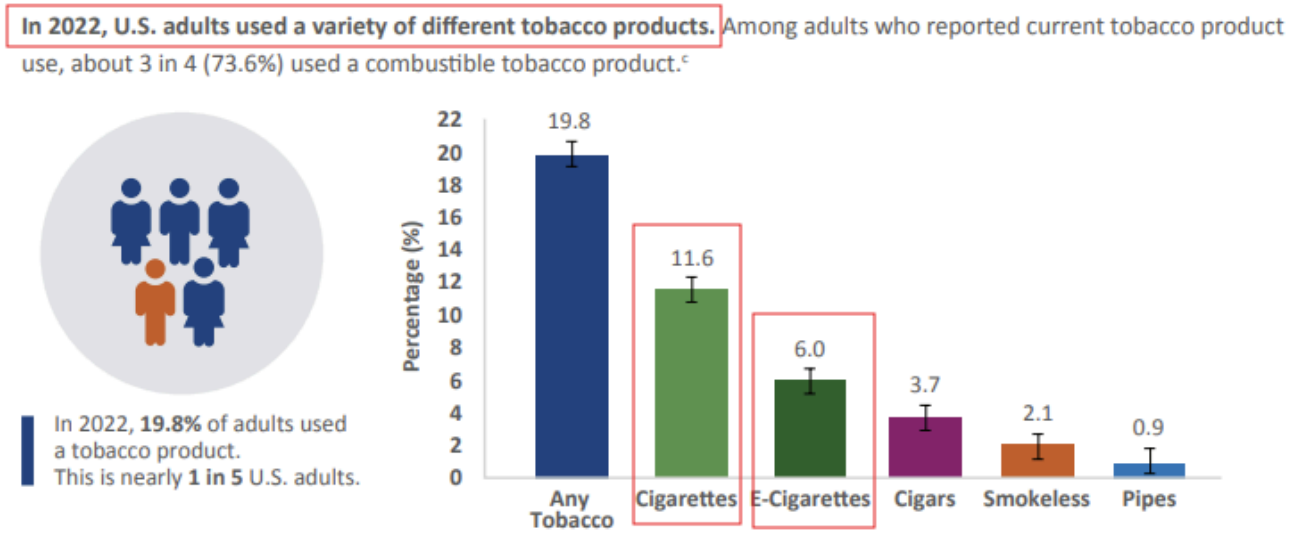


Figure 3. Tobacco product use among adults, United States, 2022 (Credit: CDC).

Group	E-cigarettes (S, I, R)	Cigarettes (S, I, R)
Youth	0.80, 0.20, 0	0.917, 0.083, 0
Young adults	0.94, 0.06, 0	0.884, 0.116, 0
Adults	0.94, 0.06, 0	0.884, 0.116, 0

2.3 Parameters

Aging. Youth spend an average of 7 years in their group (aging rate 1/84 per month) and young adults 5 years (1/60 per month).

Death rate. The CDC reports an age-adjusted death rate of 798.85 per 100,000 in 2022 (CDC Data Brief #521), or 0.0079885 per person per year. Dividing by 12 gives $\mu \approx 0.0006657$ per month.

Transmission (β). 18% of youth say they would try an e-cigarette if a friend offered one (U.S. Surgeon General, 2016), and 3% say the same about cigarettes (NSDUH, 2014). The model uses these as monthly transmission coefficients: $\beta_0 = 0.18$ for e-cigarettes and 0.003 for cigarettes. Section 4 shows that this conversion is a major source of error.

Quitting (γ). For e-cigarettes the model uses a 33.3% quit-attempt rate (Quit Attempts and Intentions, 2022). For cigarettes it uses the 67.7% of smokers who want to quit (CDC, 2022). Each annual percentage p is converted to a monthly rate as $(1 + p)^{(1/12)} - 1$, giving $\gamma_0 \approx 0.024$ for e-cigarettes and 0.044 for cigarettes.

Relapse (ρ). Annual relapse rates of 60% for e-cigarettes (PATH Study, 2019) and 91.2% for cigarettes (CDC, 2022) are converted the same way, giving $\rho_0 \approx 0.040$ and 0.055 per month.

Time variation. Each rate is its base value plus a small oscillating term:

$$\beta(t) = \beta_0 + u_t \cdot \cos(\pi t/12), \quad \gamma(t) = \gamma_0 + u_t \cdot \cos(\pi t/12), \quad \rho(t) = \rho_0 + u_t \cdot \cos(\pi t/12),$$

where t is in months and u_t is drawn uniformly from [0, 0.01] each month ([0, 0.001] for cigarette β). The cosine has a period of 24 months, and because u_t is redrawn every month the term acts as a small, irregular perturbation rather than a fixed seasonal cycle. Its size is at most about 5% of β_0 for e-cigarettes.

```python
# Function to calculate beta values - Infection
def beta_values(t):
    if product_type == 'EC':
        return 0.18 + (random.random() / 100) * np.cos(t / 12 * np.pi)
    else:
        return 0.003 + (random.random() / 1000) * np.cos(t / 12 * np.pi)

# Function to calculate gamma values - Recovery
def gamma_values(t):
    if product_type == 'EC':
        return (1.333 ** (1 / 12) - 1) + (random.random() / 100) * np.cos(t / 12 * np.pi)
    else:
        return (1.677 ** (1 / 12) - 1) + (random.random() / 100) * np.cos(t / 12 * np.pi)

# Function to calculate rho values - Relapse
def rho_values(t):
    if product_type == 'EC':
        return (1.600 ** (1 / 12) - 1) + (random.random() / 100) * np.cos(t / 12 * np.pi)
    else:
        return (1.912 ** (1 / 12) - 1) + (random.random() / 100) * np.cos(t / 12 * np.pi)
```

Figure 4. Code for the time-varying β, γ, and ρ functions (Credit: Reva Khaire).

2.4 Delays

1. **Initiation delay, τ_1 = 12 months.** Starting reflects months of accumulated exposure to peers and marketing (U.S. Surgeon General, 2016).
2. **Quitting delay, τ_2 = 6 months.** Withdrawal and adjustment take months (Urban Recovery, 2024).
3. **Relapse delay, τ_3 = 12 months.** Relapse often peaks within a year of quitting (Prochaska & DiClemente, Stages of Change).

For $t < \tau$, the delayed value is taken equal to the current value, which corresponds to a constant history before 2022.

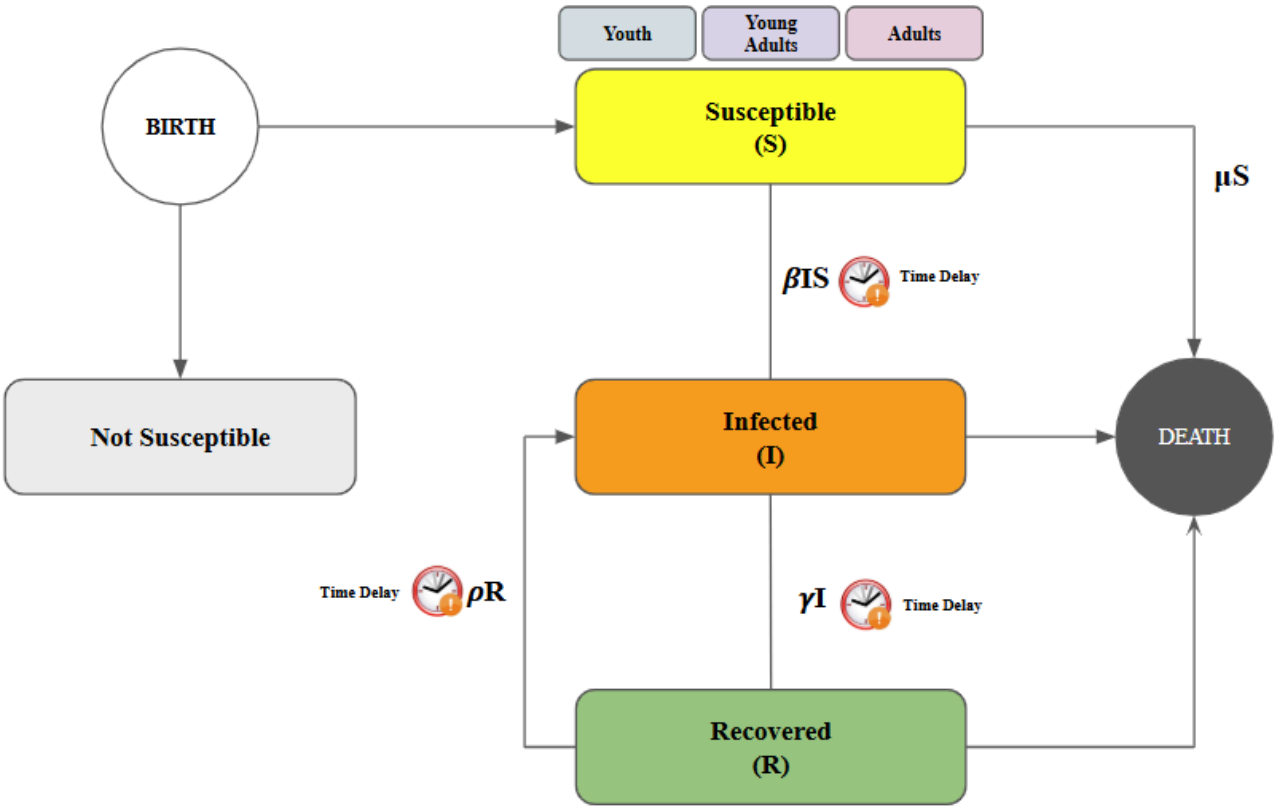


Figure 5. Structure of the delay SIR model for each age group (Credit: Reva Khaire).

2.5 Model equations

Subscripts y, ya, and a denote youth, young adults, and adults; $a_1 = 1/84$ and $a_2 = 1/60$ are the aging rates; and $X_i = X(t - \tau_i)$ is a delayed value. The equations below are exactly those in the code.

Youth

- $dS_y/dt = B - \beta(t)\cdot S_y\cdot I_y_1 - \mu\cdot S_y - a_1\cdot S_y$
- $dI_y/dt = \beta(t)\cdot S_y\cdot I_y_1 - (\mu + \gamma(t))\cdot I_y + \rho(t)\cdot R_y_3 - a_1\cdot I_y$
- $dR_y/dt = \gamma(t)\cdot I_y_2 - (\mu + \rho(t))\cdot R_y - a_1\cdot R_y$

Young adults

- $dS_ya/dt = a_1\cdot S_y - \beta(t)\cdot S_ya\cdot I_y_1 - \mu\cdot S_ya - a_2\cdot S_ya$
- $dI_ya/dt = a_1\cdot I_y + \beta(t)\cdot S_ya\cdot I_y_1 - (\mu + \gamma(t))\cdot I_ya - a_2\cdot I_ya$
- $dR_ya/dt = a_1\cdot R_y + \gamma(t)\cdot I_ya_2 - (\mu + \rho(t))\cdot R_ya_3 - a_2\cdot R_ya$

Adults

- $dS_a/dt = a_2 \cdot S_{ya} - \beta(t) \cdot S_a \cdot I_a - \mu \cdot S_a$
- $dI_a/dt = a_2 \cdot I_{ya} + \beta(t) \cdot S_a \cdot I_a - (\mu + \gamma(t)) \cdot I_{a_2}$
- $dR_a/dt = a_2 \cdot R_{ya} + \gamma(t) \cdot I_{a_2} - (\mu + \rho(t)) \cdot R_{a_3}$

Three features of this structure matter for the results:

- **Young adults start using through contact with youth users** (the term $\beta \cdot S_{ya} \cdot I_{y_1}$), while adults start through contact with other adults, with no delay.
- **No older group affects a younger one.** Youth equations depend only on youth, so the system is block lower-triangular: youth → young adults → adults.
- **Delayed inflows and current outflows are not always matched.** For example, youth leave the user state at rate $\gamma \cdot I_y(t)$ but enter the former-user state at rate $\gamma \cdot I_y(t - \tau_2)$. As a result, the total population is not exactly conserved, which is one reason the code normalizes the solution (Section 2.6).

2.6 Simulation

The system is solved with the Python package ddeint over 120 monthly steps, starting in 2022 (month 0). For the population plots, the solution at each time is divided by the sum of all nine compartments, so **plotted values are shares of the total population across all three age groups**, not shares within each group. For comparison with CDC data (Section 4), the youth user share is computed within the youth group: $I_y / (S_y + I_y + R_y)$. This ratio is unaffected by the whole-population normalization.

To find which assumptions drive the differences from CDC data, the model was rerun with one or more assumptions changed: the youth starting value, a birth rate that balances aging ($B = a_1$, so inflow matches outflow from a youth group of size 1), and a scaled transmission rate.

3. Results: Population Dynamics (Original Assumptions)

Figures 6–9 show the model with the assumptions of Section 2, before the corrections of Section 4. All values are shares of the total population across the three age groups (Section 2.6). Because the youth group shrinks over time when births do not balance aging, part of every trend reflects people moving between age groups.

3.1 E-cigarettes

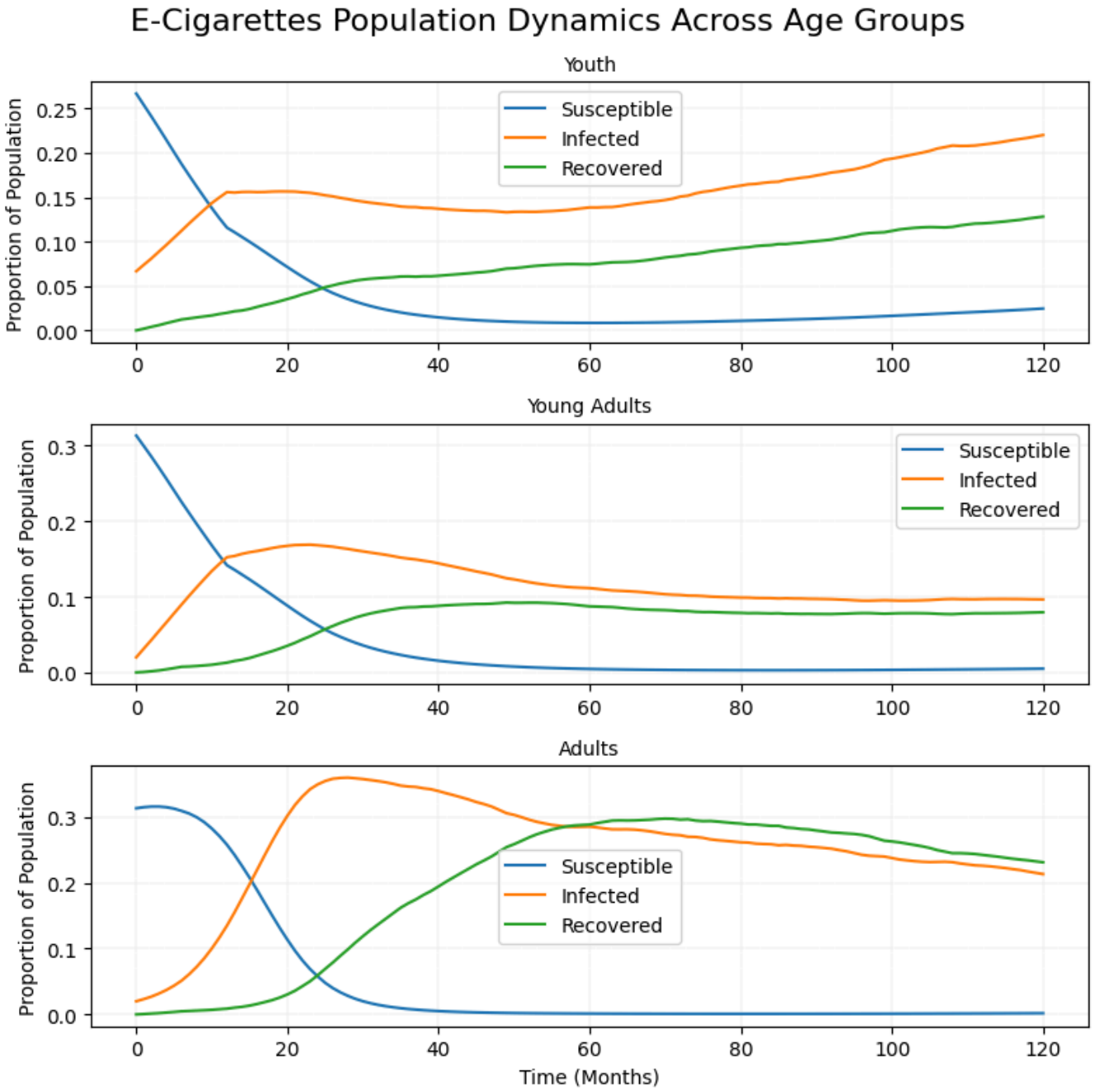


Figure 6. E-cigarette population dynamics by age group (Credit: Reva Khaire).

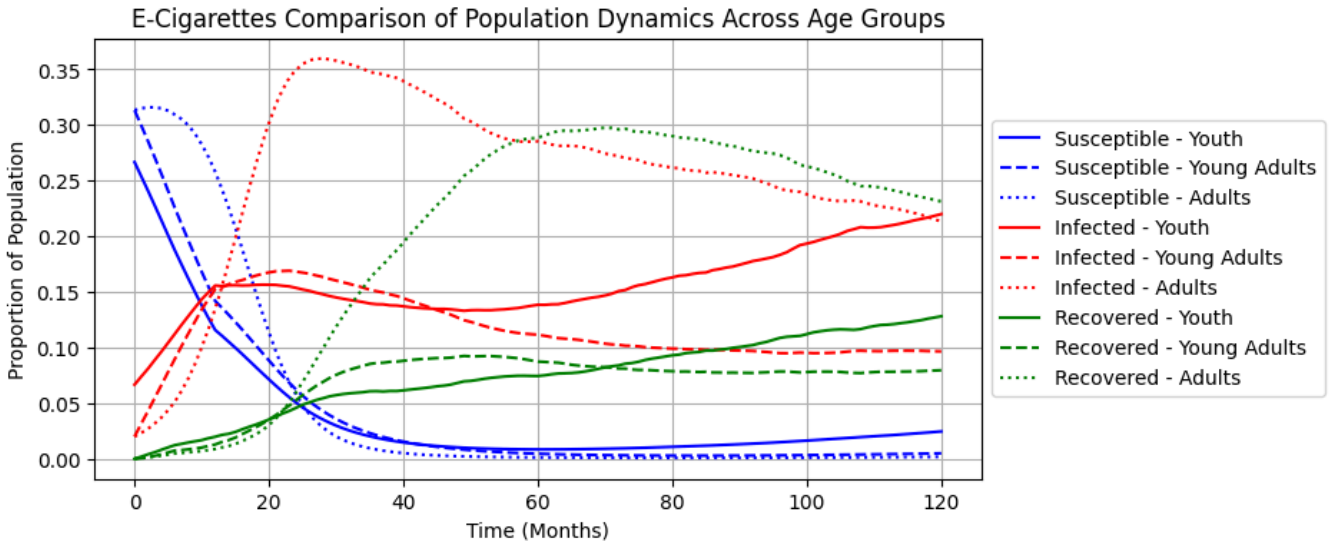


Figure 7. E-cigarette dynamics, all age groups on one plot. Values are shares of the total population (Credit: Reva Khaire).

- **Youth.** Susceptible youth fall from about 0.27 of the total population to below 0.02 within about 40 months, so almost every youth becomes a user or former user. Youth users rise from about 0.07 to about 0.16 by month 15, dip slightly, and then rise again to about 0.22 by month 120.
- **Young adults.** Users peak at about 0.17 around month 22 and then decline to about 0.10.
- **Adults.** Users rise from about 0.02 to a peak of about 0.36 near month 28, as earlier cohorts age in and adult transmission accelerates, and then decline to about 0.22 by month 120. Former adult users rise to about 0.30 by month 70.

The near-complete uptake among youth is not realistic. It is the first sign that the original assumptions overstate transmission, which Section 4 confirms.

3.2 Cigarettes

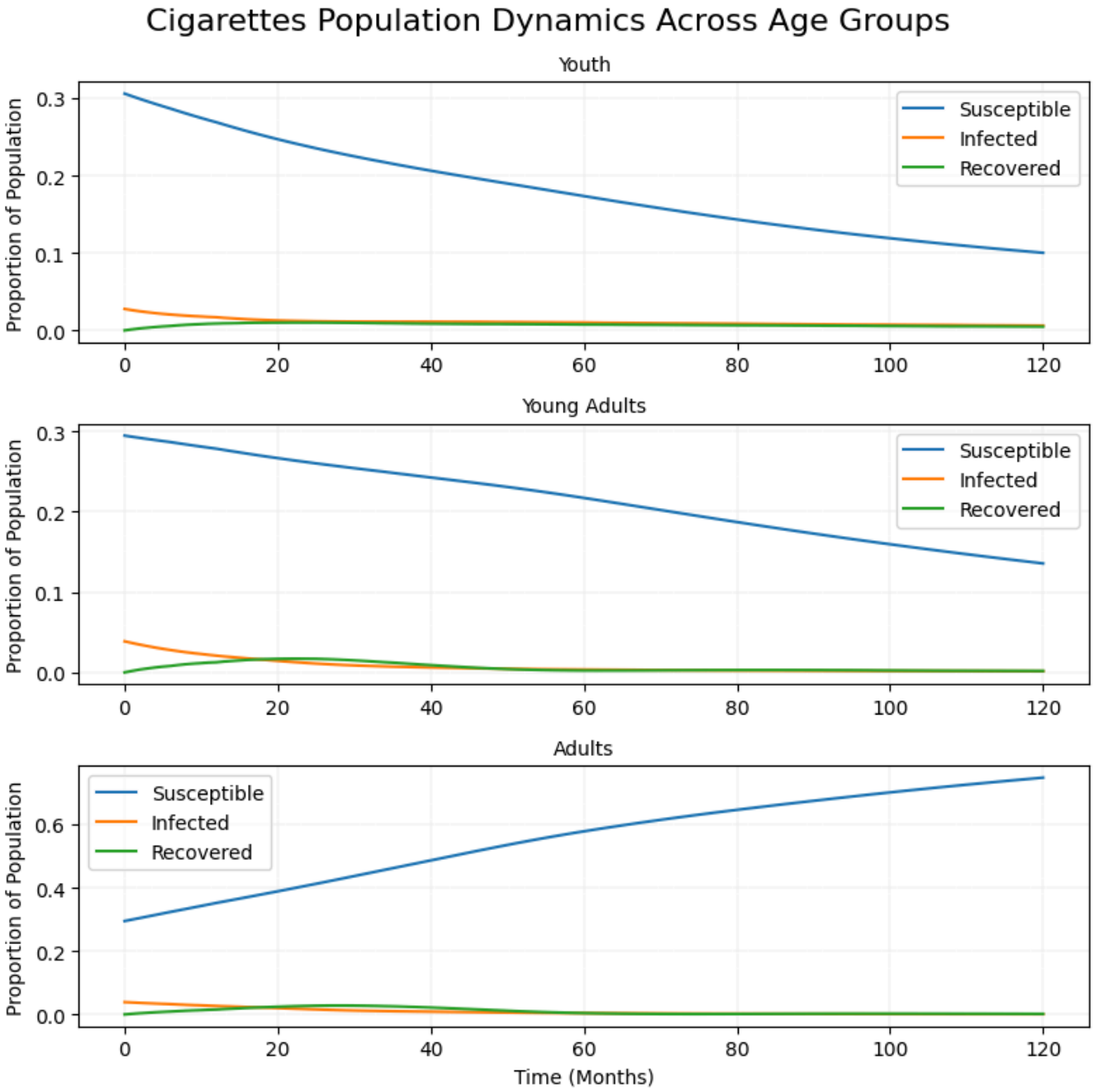


Figure 8. Cigarette population dynamics by age group (Credit: Reva Khaire).

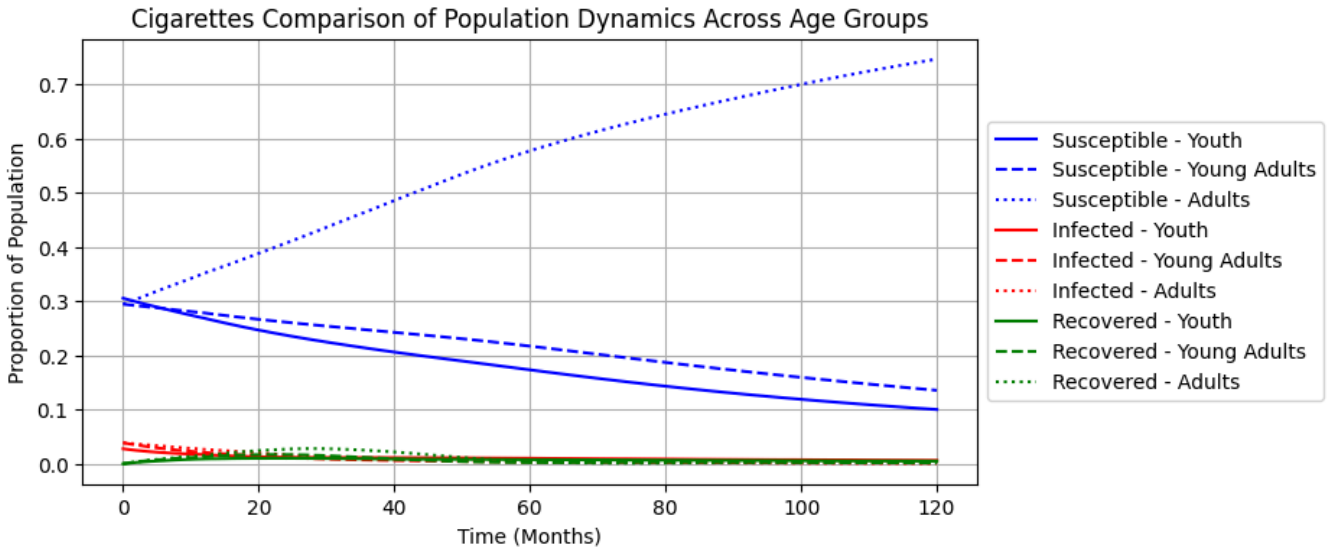


Figure 9. Cigarette dynamics, all age groups on one plot. Values are shares of the total population (Credit: Reva Khaire).

- **Users and former users** stay below about 0.04 of the total population in every group and decline toward zero over the 10 years. With $\beta_0 = 0.003$, transmission is too weak to sustain cigarette use, and quitting removes users faster than new ones start.
- **Susceptible youth** fall from about 0.30 to about 0.10, and **susceptible young adults** from about 0.29 to about 0.14. **Susceptible adults** rise from about 0.30 to about 0.75. These changes come mainly from aging: youth and young adults move into the adult group faster than births replace them, rather than from people starting to smoke.

4. Comparison with CDC Data

4.1 Correcting the comparison

The earlier version of this paper reported a 2023 prediction of 6.67% current e-cigarette use among youth, close to the CDC's 7.7%. That number came from two errors in the comparison code. It read the model's value at month 0 (the starting value) instead of month 12, and it divided by the total of all nine compartments, which reduced the youth share by about a factor of three (20% ÷ 3 ≈ 6.67%). The "match" was therefore the starting value, not a prediction: changing β, or removing all delays, gave the same 6.67%.

The corrected comparison computes the youth user share within the youth group, I_y / (S_y + I_y + R_y), at months 12 (2023) and 24 (2024), and compares it with current use from the National Youth Tobacco Survey.

4.2 E-cigarettes

	2023	2024
Model (original assumptions)	53.5%	61.5%
CDC, current use	7.7%	5.9%

The model overpredicts youth e-cigarette use by a factor of about 7 in 2023 and 10 in 2024. Changing one assumption at a time shows where the gap comes from:

Scenario	Youth group size, 2024	2023	2024
As written	0.72	53.5%	61.5%
Start from current use (9.4%)	0.72	34.4%	45.9%
Births balance aging	0.93	51.4%	57.7%
Transmission rate ÷ 10	0.76	18.3%	17.3%
All three	0.98	7.8%	6.8%
CDC		7.7%	5.9%

1. **Starting value.** The 20% starting value measures ever-use. Current use in 2022 was 9.4% of middle and high school students combined (Cooper et al., 2022).
2. **Population balance.** With B = 0.00103 per month and an aging rate of 1/84 per month, far fewer youth enter than leave, and the youth group shrinks to 72% of its starting size in two years. With births balancing aging, it stays near its original size.
3. **Transmission rate.** The 18% figure is the share of youth who would accept an offer, not the monthly rate at which susceptible youth start using. Using it directly as β makes transmission far too fast.

The transmission rate has by far the largest effect. With all three corrections, the model comes close to the CDC values in both years: 7.8% vs. 7.7% in 2023 and 6.8% vs. 5.9% in 2024.

To test whether any constant transmission rate fits both years, β was scaled while keeping the corrected starting value and birth rate:

β scale	2023	2024
1.00	33.3%	42.9%
0.50	16.0%	19.6%
0.25	10.4%	10.9%
0.15	8.6%	8.1%
0.10	7.8%	6.8%
0.05	7.1%	5.7%
CDC	7.7%	5.9%

No single scale matches both years. A scale near 0.10 matches 2023 but overpredicts 2024; a scale near 0.05 comes close to 2024 but underpredicts 2023. With one free parameter and two observations, the model could have failed this test, and it did. The decline the CDC observed between 2023 and 2024 is steeper than a constant-rate model with these other assumptions produces. Possible explanations include a transmission rate that fell over time (for example, because of policy or enforcement changes), faster quitting than assumed, or remaining errors in the starting values. The available data do not distinguish among them.

4.3 Cigarettes

Scenario	Youth group size, 2024	2023	2024
As written	0.76	5.8%	4.6%
Start from current use (1.6%)	0.76	1.1%	0.9%
Births balance aging	0.99	5.1%	3.6%
Both	0.99	1.0%	0.7%
CDC		1.6%	1.4%

With both corrections, the model predicts 1.0% (2023) and 0.7% (2024), below the CDC's 1.6% and 1.4%. Scaling the transmission rate barely changes the result:

Scale	β: 2023	β: 2024	γ: 2023	γ: 2024
10	1.3%	1.2%		
5	1.1%	0.9%		
2	1.0%	0.7%	0.7%	0.5%
1	1.0%	0.7%	1.0%	0.7%
0.5	1.0%	0.7%	1.2%	0.9%
0.25			1.3%	1.1%
CDC	1.6%	1.4%	1.6%	1.4%

Even a tenfold increase in β moves the 2023 prediction only from 1.0% to 1.3%. The cigarette predictions are controlled mainly by the starting value and the quit rate. The quit rate used (based on 67.7% of smokers *wanting* to quit) is likely too high, since wanting to quit is not the same as quitting; reducing γ brings the predictions closer to the CDC values.

4.4 What the data can and cannot determine

In the equations, transmission appears only as β·S·I. If β is understood as (how often a susceptible teen encounters vaping) × (the probability of accepting), only the product enters the model. Doubling one factor and halving the other leaves every prediction unchanged. Prevalence data, however many years are available, therefore cannot identify the two factors separately; this is a structural identifiability problem.

Similarly, the two years of prevalence data cannot tell whether the gap in 2024 comes from initiation, quitting, or the starting assumptions, since each can produce similar prevalence curves. Separating initiation from quitting would require data on how many teens start each year (incidence), and separating encounter frequency from acceptance would require measuring one of them directly.

5. Sensitivity Analysis

5.1 Setup

The sensitivity analysis varies β, γ, and ρ one at a time from 90% to 110% of their base values (five levels each). It uses a simplified version of the model that differs from the main model in four ways:

- **No delays.** Every term uses current values, so this version is a system of ordinary differential equations.
- **Young adults start through contact with other young adults** (β·S_ya·I_ya), not youth.
- **Transmission declines over time:** $\beta(t) = \beta_0 \cdot e^{(-0.25t/12)} + 0.005 \cdot \cos(\pi t/12)$, so β falls by about 22% per year. γ and ρ are constant.
- **Both products use the e-cigarette starting values** (youth 0.80, 0.20, 0; young adults and adults 0.94, 0.06, 0). *[Reva: the cigarette run should use the cigarette starting values; fix the history function in the cigarette sensitivity cell and rerun, or keep this note.]*

The results therefore show how this simplified model responds to each rate. They should not be read as the sensitivity of the delay model in Sections 3 and 4. Values are shares of the total population.

5.2 E-cigarettes

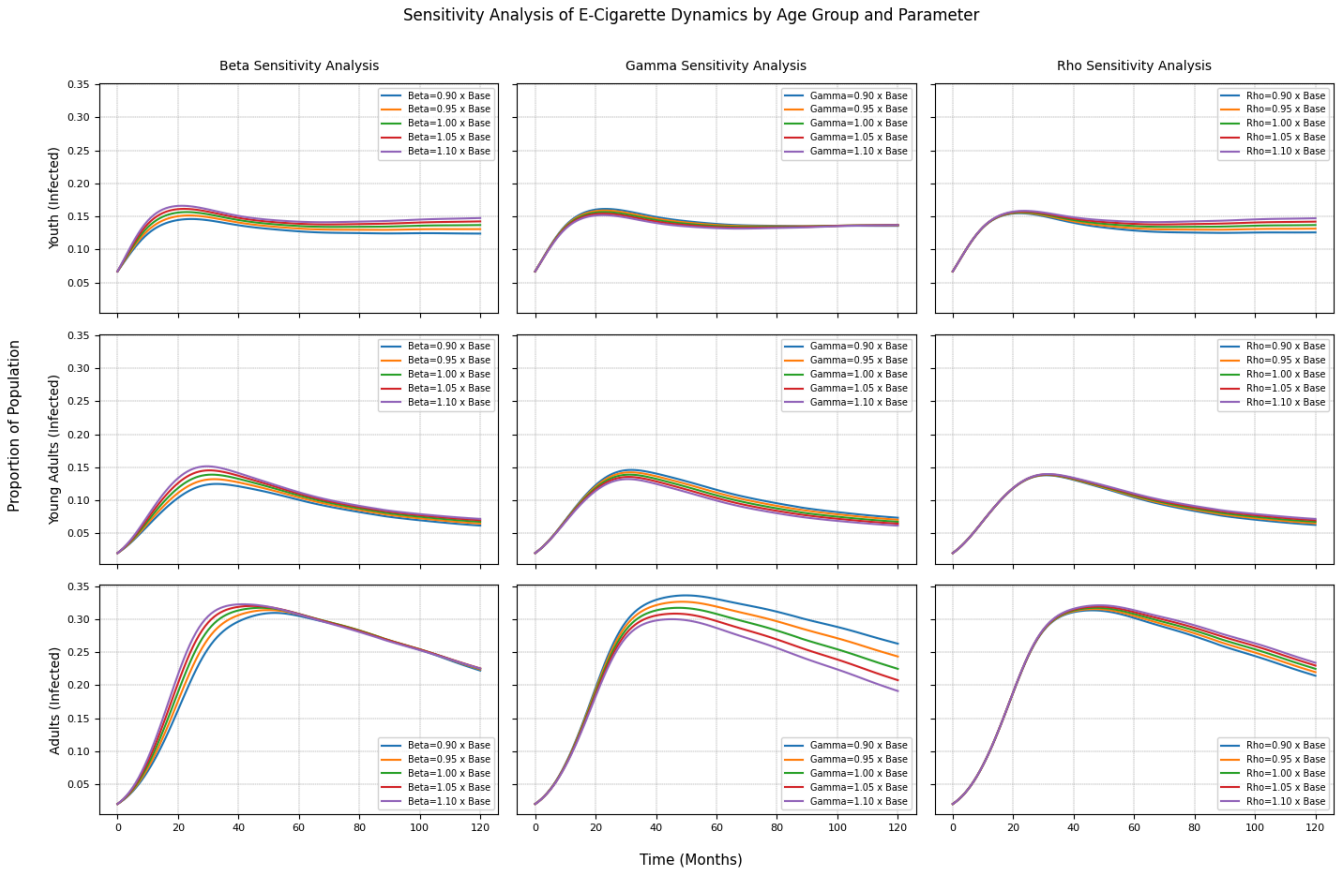


Figure 10. Sensitivity of e-cigarette use to β, γ, and ρ by age group, simplified model (Credit: Reva Khaire).

- **Transmission (β)** has the largest effect on early growth. Higher β gives earlier and higher peaks: youth users peak at about 0.15 of the total population around month 20, young adults at about 0.14 around month 30, and adults at about 0.32 around month 45.
- **Quitting (γ)** mainly affects the decline after the peak. Its effect is largest for adults: at month 120, adult users range from about 0.19 (γ × 1.1) to about 0.26 (γ × 0.9).
- **Relapse (ρ)** mainly affects the later plateau, with almost no effect on early growth.

5.3 Cigarettes

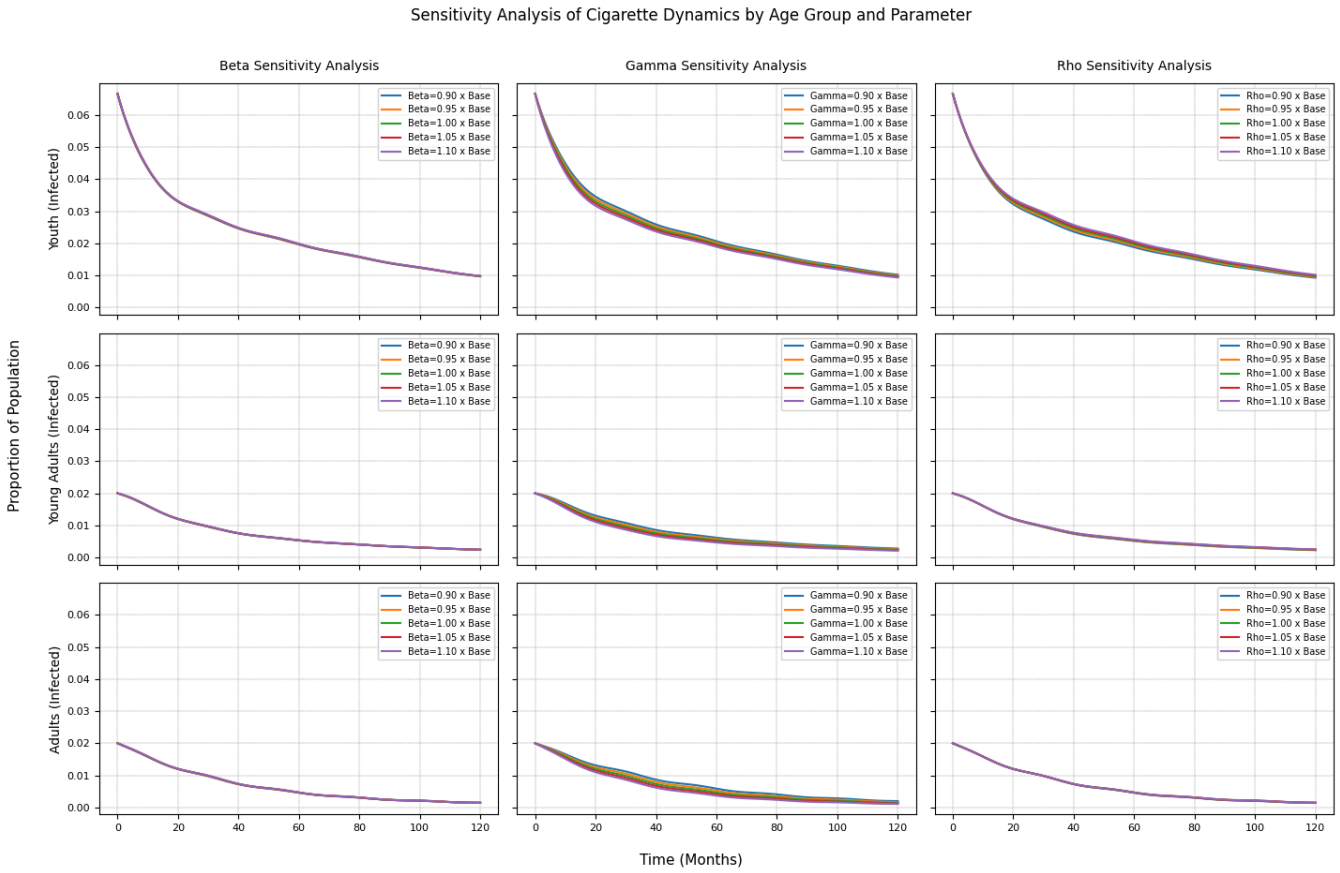


Figure 11. Sensitivity of cigarette use to β, γ, and ρ by age group, simplified model with e-cigarette starting values (Credit: Reva Khaire).

Use declines in every age group under all settings, and ±10% changes in β make no visible difference. Changes in γ and ρ have small effects. With $\beta_0 = 0.003$, transmission is too weak to matter, so cigarette use is governed by quitting and by the starting values. This agrees with the corrected comparison in Section 4.3, where even a tenfold change in β barely moved the prediction.

6. Conclusion

- **The original assumptions greatly overpredict youth e-cigarette use.** The model as written predicts 53.5% current use in 2023 and 61.5% in 2024, against CDC values of 7.7% and 5.9%. Nearly all susceptible youth become users within about three years, which is not realistic.
- **Three assumptions explain most of the gap:** an ever-use starting value, a birth rate that does not balance aging, and treating the 18% offer-acceptance figure as a monthly transmission rate. The transmission rate has the largest effect. With all three

corrected, the model comes close to the CDC values in both years (7.8% vs. 7.7% in 2023, 6.8% vs. 5.9% in 2024), down from 53.5% and 61.5% before the corrections.

- **No single transmission rate matches both years exactly.** The observed decline from 2023 to 2024 is steeper than the model produces, which suggests that a rate changed over time or that another assumption is still wrong.
- **Prevalence data cannot identify every parameter.** Contact frequency and acceptance probability enter only as a product, and two years of prevalence data cannot separate initiation, quitting, and starting assumptions.
- **Cigarette predictions depend on the starting value and the quit rate,** not on transmission. A tenfold change in β barely changes them.

7. Limitations

- **Parameter sources.** The quit rate for cigarettes is based on the share of smokers who *want* to quit, which overstates actual quitting. The conversion of annual percentages to monthly rates, (1 + p)^(1/12) − 1, is an approximation.
- **Delayed terms are not always matched.** Some transitions use a delayed value for the inflow and a current value for the outflow (Section 2.5), so the total population is not exactly conserved. In the young-adult and adult former-user equations, deaths are applied to the delayed value $R(t - \tau_3)$ rather than the current value.
- **Time variation.** The cosine term has a 24-month period, and its amplitude is redrawn randomly each month, so it does not represent a fixed seasonal cycle.
- **Age structure.** Young adults (ages 18–23) are assigned an average stay of 5 years. Young adults start using only through contact with youth users, and adults do not influence youth, although in reality they do.
- **Sensitivity analysis.** It uses a simplified model without delays and with a declining transmission rate (Section 5.1), so it does not measure the sensitivity of the delay model directly.
- **Validation data.** Only two years of CDC data were used.

8. Future Work

- **Stability.** Linearize the model near an equilibrium and derive its characteristic equation. Because the age groups affect each other in only one direction, the characteristic determinant factors into one block per age group. The delays add factors

of $e^{(-\lambda\tau)}$, and an open question is whether they can make the system oscillate or change its stability.

- **Identifiability.** Determine which parameters, including the delays, can be recovered from prevalence data, and what additional data (such as yearly initiation counts) would separate the remaining ones.
- **Time-varying transmission.** Fit a transmission rate that changes over time to a longer series of CDC data and test whether it explains the 2023–2024 decline.
- **Model corrections.** Match the delayed inflow and outflow terms, use within-group shares in all plots, and rerun the sensitivity analysis on the delay model with the correct starting values for each product.
- **Dual use.** Add transitions between e-cigarettes and cigarettes.

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